

TUGAS
Komputasi Tersebar dan Paralel



JUDUL
INSTALASI APACHE KAFKA & JAVA RMI

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RINGKASAN

Laporan ini disusun untuk memenuhi tugas instalasi dan demonstrasi Apache Kafka serta implementasi Java RMI (*Remote Method Invocation*). Apache Kafka digunakan sebagai *message broker* untuk mendukung komunikasi *asynchronous* dan pemrosesan data secara *real-time*, sedangkan Java RMI digunakan untuk mendemonstrasikan pemanggilan *method* antar objek Java yang berjalan pada JVM berbeda secara *synchronous*.

Pada bagian Apache Kafka, instalasi dilakukan dengan menggunakan dua pendekatan :

1. Pendekatan Manual (*Native*): Dilakukan instalasi dan konfigurasi langsung pada sistem operasi Windows menggunakan *Command Prompt* (CMD). Pada metode ini, diterapkan mode terbaru KRaft (Kafka Raft) yang beroperasi tanpa Zookeeper. Pengujian manual via CMD mencakup konfigurasi *server.properties*, formatting storage ID, serta simulasi pengiriman pesan (*Producer*) dan penerimaan pesan (*Consumer*) secara langsung melalui terminal.
2. Pendekatan Berbasis Kontainer: Menggunakan Docker dan Docker Compose untuk menjalankan *service* Kafka dalam lingkungan yang terisolasi dan mudah dikelola.

Pada bagian Java RMI, dibuat aplikasi sederhana yang terdiri dari *interface remote*, *server*, dan *client*. *Server* menyediakan *method* yang dapat dipanggil secara *remote* oleh *client*. Hasil pengujian menunjukkan bahwa *client* berhasil menerima *respons* dari *server*, membuktikan bahwa mekanisme Java RMI berjalan dengan baik.

1. Install Apache Kafka - CMD

Pada pendekatan manual ini, instalasi dilakukan menggunakan mode KRaft (tanpa Zookeeper). Seluruh perintah dijalankan melalui *Command Prompt* (CMD) pada direktori instalasi C:\kafka. Berikut adalah langkah-langkah teknis beserta *script* perintah yang dieksekusi:

Langkah 1: Inisialisasi Cluster ID (UUID)

Langkah pertama adalah membuat identitas unik untuk klaster Kafka. Perintah berikut dijalankan untuk men-*generate* kode UUID secara acak dengan `.\bin\windows\kafka-storage.bat random-uuid`

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7171]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Naila>java --version
openjdk 17.0.16 2025-07-15 LTS
OpenJDK Runtime Environment Microsoft-11926163 (build 17.0.16+8-LTS)
OpenJDK 64-Bit Server VM Microsoft-11926163 (build 17.0.16+8-LTS, mixed mode, sharing)

C:\Users\Naila>
```

Langkah 2: Formatting Log Storage

Menggunakan UUID yang telah didapatkan, dilakukan proses *formatting* pada direktori penyimpanan log Kafka agar *server* mengenali klaster tersebut. *Script* yang dijalankan adalah

```
.\bin\windows\kafka-storage.bat format -t UUID -c
.\config\kraft\server.properties
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Naila> java --version
openjdk 17.0.16 2025-07-15 LTS
OpenJDK Runtime Environment Microsoft-11926163 (build 17.0.16+8-LTS)
OpenJDK 64-Bit Server VM Microsoft-11926163 (build 17.0.16+8-LTS, mixed mode, sharing)
PS C:\Users\Naila> cd C:\kafka
PS C:\kafka> .\bin\windows\kafka-storage.bat random-uuid
2025-12-14T12:17:55.157831400Z main ERROR Reconfiguration failed: No configuration found for '5e2de80c' at 'null' in 'null'
fstWjkkxkQKGr_Q5RomzEzQ
PS C:\kafka> .\bin\windows\kafka-storage.bat format -t fstWjkkxkQKGr_Q5RomzEzQ -c .\config\kraft\server.properties
2025-12-14T12:18:20.070312900Z main ERROR Reconfiguration failed: No configuration found for '5e2de80c' at 'null' in 'null'
Formatting metadata directory C:\kafka\kafka_2.13-4.1.1\data-kraft with metadata.version 4.1-IV1.
PS C:\kafka>
```

Langkah 3: Menjalankan Server Kafka (Broker)

Setelah konfigurasi penyimpanan selesai, *server* Kafka dijalankan menggunakan konfigurasi KRaft.

Perintah `.\bin\windows\kafka-server-start.bat`

.\config\kraft\server.properties akan menjalankan *service* Kafka sebagai proses utama pada terminal tersebut. Tunggu sampai muncul “kafka server started”

```
Windows PowerShell
ssl.secure.random.implementation = null
ssl.trustmanager.algorithm = PKIX
ssl.truststore.certificates = null
ssl.truststore.location = null
ssl.truststore.password = null
ssl.truststore.type = JKS
telemetry.max.bytes = 1048576
transaction.abort.timed.out.transaction.cleanup.interval.ms = 10000
transaction.max.timeout.ms = 900000
transaction.partition.verification.enable = true
transaction.remove.expired.transaction.cleanup.interval.ms = 3600000
transaction.state.log.load.buffer.size = 5242880
transaction.state.log.min.isr = 1
transaction.state.log.num.partitions = 50
transaction.state.log.replication.factor = 1
transaction.state.log.segment.bytes = 104857600
transaction.two.phase.commit.enable = false
transactional.id.expiration.ms = 604800000
unclean.leader.election.enable = false
unclean.leader.election.interval.ms = 300000
unstable.api.versions.enable = false
unstable.feature.versions.enable = false
(org.apache.kafka.common.config.AbstractConfig)
[2025-12-14 19:19:37,405] INFO [BrokerServer id=1] Waiting for the broker to be unfenced (kafka.server.BrokerServer)
[2025-12-14 19:19:37,454] INFO [BrokerLifecycleManager id=1] The broker has been unfenced. Transitioning from RECOVERY to RUNNING. (kafka.server.BrokerLifecycleManager)
[2025-12-14 19:19:37,454] INFO [BrokerServer id=1] Finished waiting for the broker to be unfenced (kafka.server.BrokerServer)
[2025-12-14 19:19:37,456] INFO [BrokerServer id=1] authorizerStart completed for endpoint PLAINTEXT. Endpoint is now READY. (org.apache.kafka.server.network.EndpointReadyFuture)
[2025-12-14 19:19:37,456] INFO [SocketServer listenerType=BROKER, nodeId=1] Enabling request processing. (kafka.network.SocketServer)
[2025-12-14 19:19:37,458] INFO [BrokerServer id=1] Waiting for all of the SocketServer Acceptors to be started (kafka.server.BrokerServer)
[2025-12-14 19:19:37,458] INFO [BrokerServer id=1] Finished waiting for all of the authorizer futures to be completed (kafka.server.BrokerServer)
[2025-12-14 19:19:37,458] INFO [BrokerServer id=1] Finished waiting for all of the SocketServer Acceptors to be started (kafka.server.BrokerServer)
[2025-12-14 19:19:37,460] INFO [BrokerServer id=1] Finished waiting for all of the SocketServer Acceptors to be started (kafka.server.BrokerServer)
[2025-12-14 19:19:37,460] INFO [BrokerServer id=1] Transition from STARTING to STARTED (kafka.server.BrokerServer)
[2025-12-14 19:19:37,462] INFO [Kafka version: 4.1.1 (org.apache.kafka.common.utils.AppInfoParser)
[2025-12-14 19:19:37,462] INFO [Kafka commitId: be816b82d25370ce (org.apache.kafka.common.utils.AppInfoParser)
[2025-12-14 19:19:37,462] INFO [Kafka startTimeMs: 1765714777460 (org.apache.kafka.common.utils.AppInfoParser)
[2025-12-14 19:19:37,462] INFO [KafkaRaftServer nodeId=1] Kafka Server started (kafka.server.KafkaRaftServer)
```

Langkah 4: Membuat Topic Baru

Pada jendela terminal baru (CMD 2), dilakukan pembuatan *topic* sebagai wadah penyimpanan pesan. *Topic* ini diberi nama tes-kafka dan dikonfigurasi untuk berjalan pada *local server* (port 9092) menggunakan *script* .\bin\windows\kafka-topics.bat --create --topic tes-kafka --bootstrap-server localhost:9092

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Naila> cd C:/kafka
PS C:\kafka> .\bin\windows\kafka-topics.bat --create --topic tes-kafka --bootstrap-server localhost:9092
2025-12-14T12:21:45.741201800Z main ERROR Reconfiguration failed: No configuration found for '5e2de80c' at 'null' in 'null'
Created topic tes-kafka.
```

Langkah 5: Mengirim Pesan (Producer)

Di terminal yang sama (CMD 2), dijalankan *script* untuk mengaktifkan mode *Producer*. Pada tahap ini, input teks yang diketikkan akan dikirimkan ke dalam *topic* tes-kafka:

.\bin\windows\kafka-console-producer.bat --topic tes-kafka --bootstrap-server localhost:9092

```
Windows PowerShell x Windows PowerShell x + v
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Naila> cd C:/kafka
PS C:\kafka> .\bin\windows\kafka-topics.bat --create --topic tes-kafka --bootstrap-server localhost:9092
2025-12-14T12:21:45.741201800Z main ERROR Reconfiguration failed: No configuration found for '5e2de80c' at 'null' in 'null'
Created topic tes-kafka.
PS C:\kafka> .\bin\windows\kafka-console-producer.bat --topic tes-kafka --bootstrap-server localhost:9092
2025-12-14T12:22:02.543365200Z main ERROR Reconfiguration failed: No configuration found for '5e2de80c' at 'null' in 'null'
>Tes Kafka
>Tes Kafka 2
>Tugas Instalasi Kafka
>|
```

Langkah 6: Menerima Pesan (*Consumer*)

Pada jendela terminal ketiga (CMD 3), dijalankan *script* untuk mengaktifkan mode *Consumer*. Parameter `--from-beginning` digunakan untuk memastikan seluruh pesan yang ada di dalam *topic* tes-kafka terbaca sejak awal:

```
.\bin\windows\kafka-console-consumer.bat --topic tes-kafka
--from-beginning --bootstrap-server localhost:9092
```

```
Windows PowerShell x Windows PowerShell x + v - □ × Windows PowerShell x + v - □ ×
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Naila> cd C:/kafka
PS C:\kafka> .\bin\windows\kafka-topics.bat --create --topic tes-kafka --bootstrap-server localhost:9092
2025-12-14T12:21:45.741201800Z main ERROR Reconfiguration failed: No configuration found for '5e2de80c' at 'null' in 'null'
Created topic tes-kafka.
PS C:\kafka> .\bin\windows\kafka-console-producer.bat --topic tes-kafka --bootstrap-server localhost:9092
2025-12-14T12:22:02.543365200Z main ERROR Reconfiguration failed: No configuration found for '5e2de80c' at 'null' in 'null'
>Tes Kafka
>Tes Kafka 2
>Tugas Instalasi Kafka
>|

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

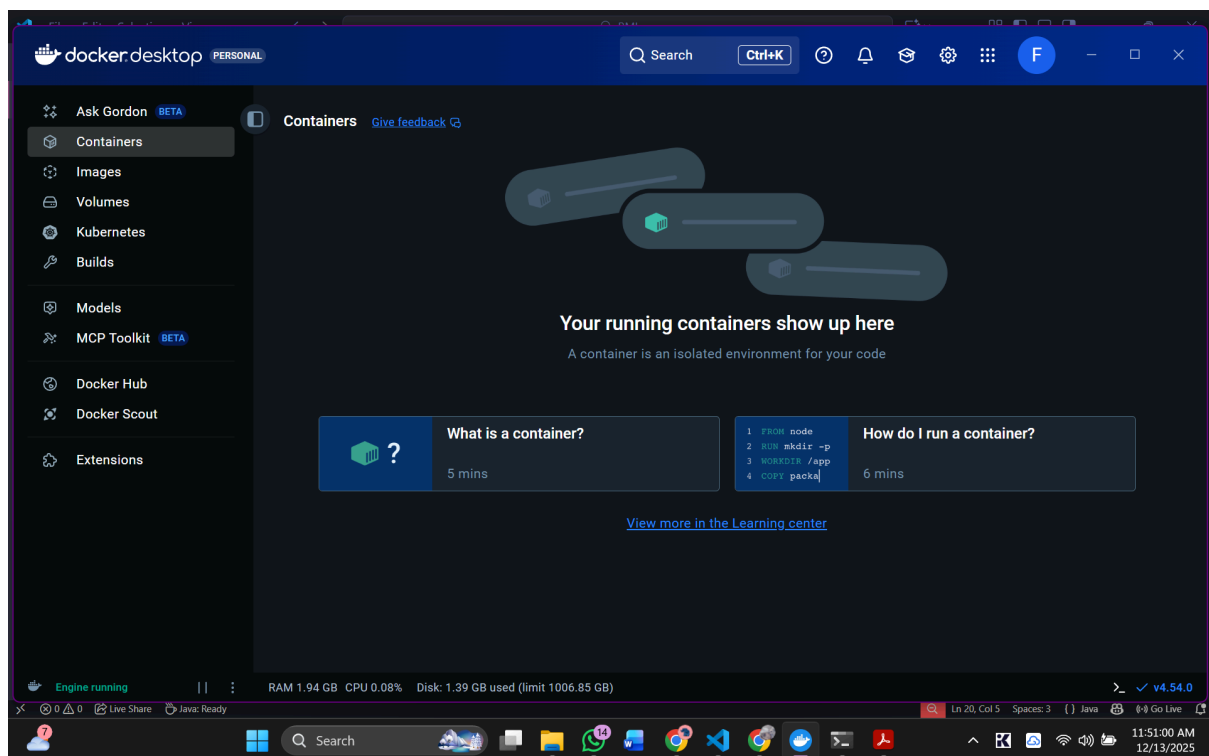
cd C:/kafka
PS C:\kafka> .\bin\windows\kafka-console-consumer.bat --topic tes-kafka --from-beginning --bootstrap-server localhost:9092
2025-12-14T12:22:48.420164800Z main ERROR Reconfiguration failed: No configuration found for '5e2de80c' at 'null' in 'null'
Tes Kafka
Tes Kafka 2
Tugas Instalasi Kafka
|
```

2. Install Apache Kafka - Docker

Langkah 1: Menyiapkan Docker Desktop

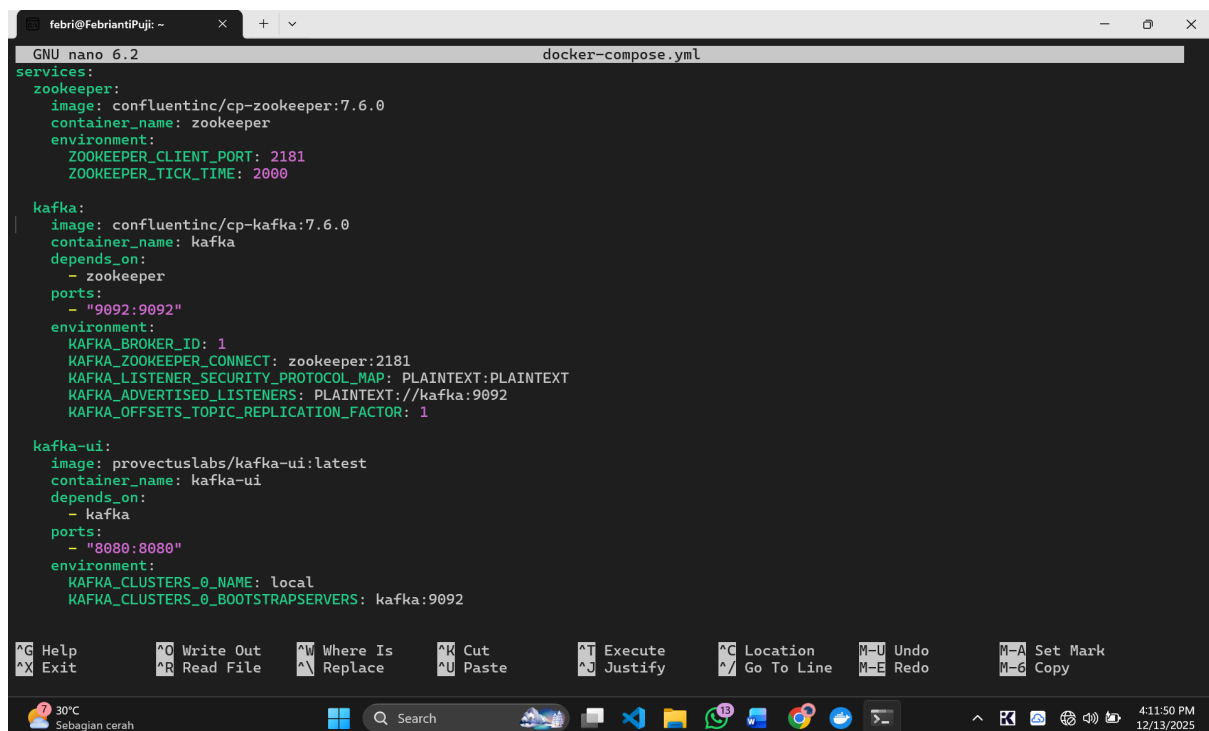
Pada tahap ini, Docker Desktop dipastikan telah terinstal dan berjalan. Docker digunakan

untuk memudahkan proses instalasi Kafka tanpa perlu konfigurasi manual yang kompleks.



Langkah 2: Menyiapkan file `docker-compose.yml`

File `docker-compose.yml` digunakan untuk mendefinisikan *service* Zookeeper dan Kafka Broker. Zookeeper berfungsi untuk manajemen metadata Kafka, sedangkan Kafka berfungsi sebagai *message broker*.



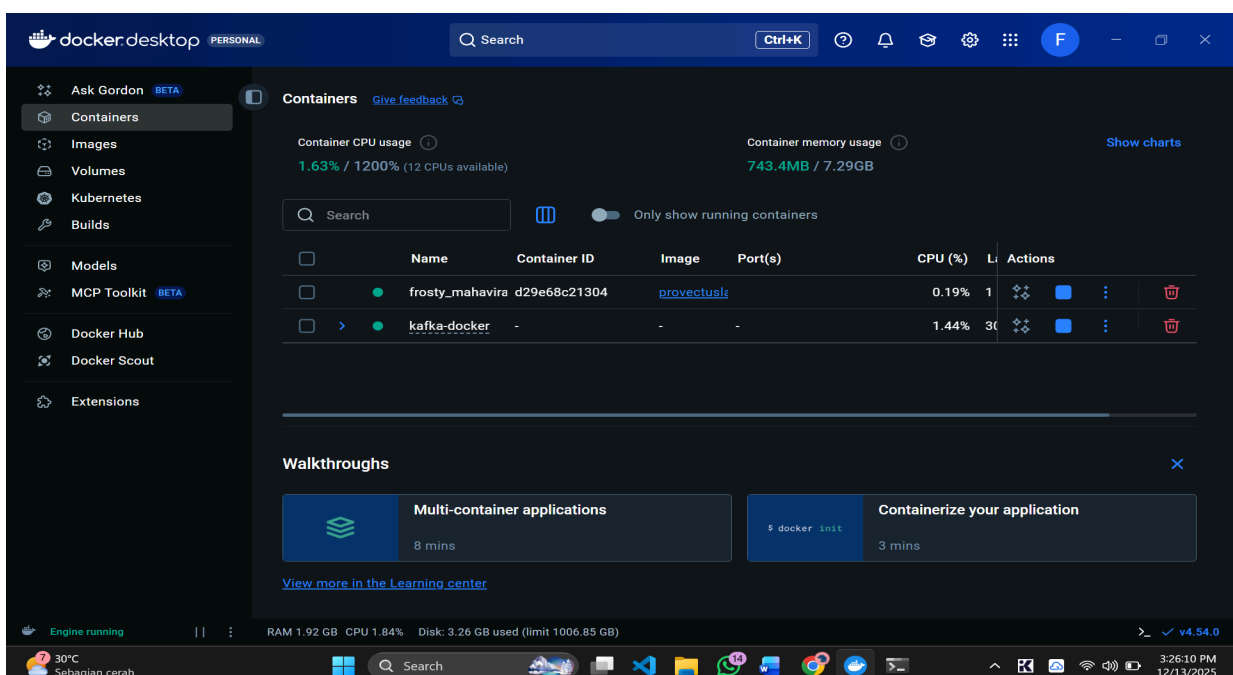
Langkah 3: Menjalankan Kafka

Perintah `docker compose up -d` dijalankan untuk menyalakan container Kafka dan Zookeeper. Status container dicek melalui Docker Desktop dan perintah `docker ps`.

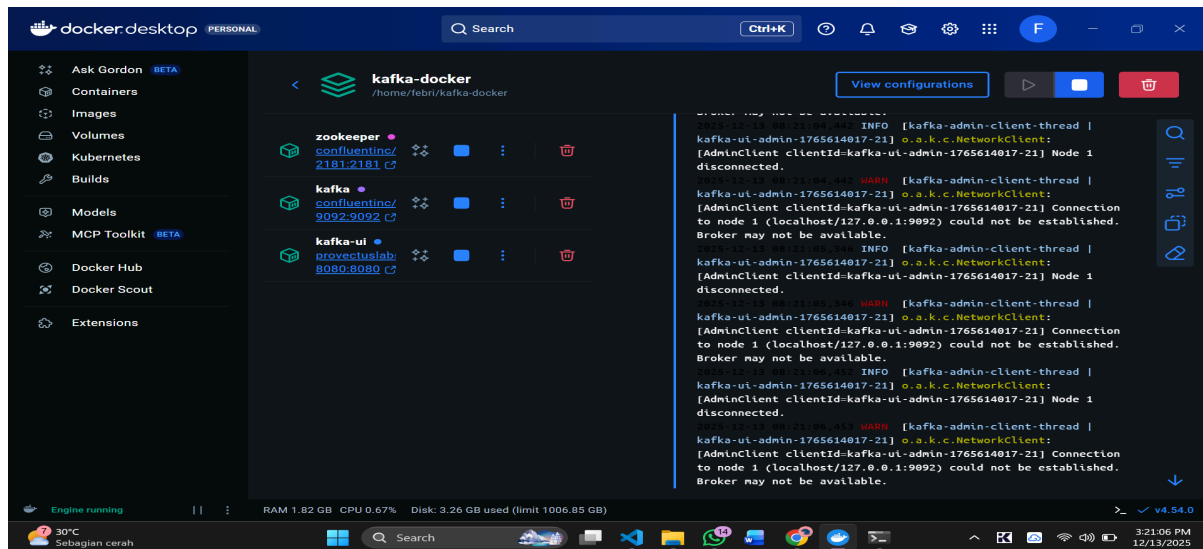
```
febri@FebriantiPuji:~$ docker compose down
docker rm -f kafka zookeeper kafka-ui
[+] Running 1/1
  ✓ Network febri_default Removed                                0.3s
kafka
zookeeper
kafka-ui
febri@FebriantiPuji:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
febri@FebriantiPuji:~$ nano docker-compose.yml
febri@FebriantiPuji:~$ docker compose up -d
[+] Running 4/4
  ✓ Network febri_default Created                                0.1s
  ✓ Container zookeeper Started                                1.0s
  ✓ Container kafka Started                                    1.4s
  ✓ Container kafka-ui Started                                1.8s
febri@FebriantiPuji:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
3ae26c6bb09e   provectuslabs/kafka-ui:latest       "/bin/sh -c 'java --..." 12 seconds ago Up 10 seconds 0.0.0.0:8080->8080/tcp, [:
9c7ae7b5d69c   confluentinc/cp-kafka:7.6.0        "/etc/confluent/dock..." 12 seconds ago Up 10 seconds 0.0.0.0:9092->9092/tcp, [:
ae81a11deaf1   confluentinc/cp-zookeeper:7.6.0    "/etc/confluent/dock..." 12 seconds ago Up 11 seconds 2181/tcp, 2888/tcp, 3888/t
cp
zookeeper
febri@FebriantiPuji:~$
```

Langkah 4: Verifikasi Kafka

Kafka diverifikasi berjalan dengan baik melalui log container yang menunjukkan `kafkaServer started`.

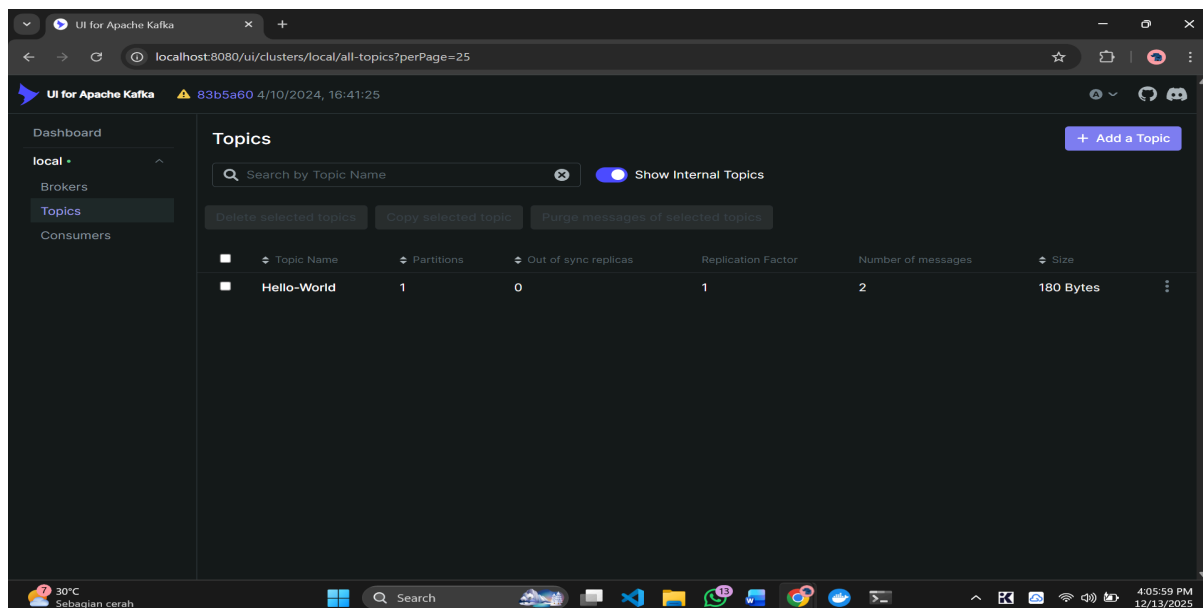


Saat container kafka-docker diklik, ditampilkan daftar *service* yang berjalan di dalamnya yaitu Zookeeper, Kafka Broker, dan Kafka UI beserta port yang digunakan. Pada sisi kanan ditampilkan log container yang menunjukkan proses komunikasi antar *service* Kafka. Pesan *warning* pada log menandakan adanya percobaan koneksi Kafka UI ke broker Kafka yang bersifat sementara dan tidak mengganggu proses utama, selama Kafka dan Zookeeper tetap dalam status berjalan (*running*).



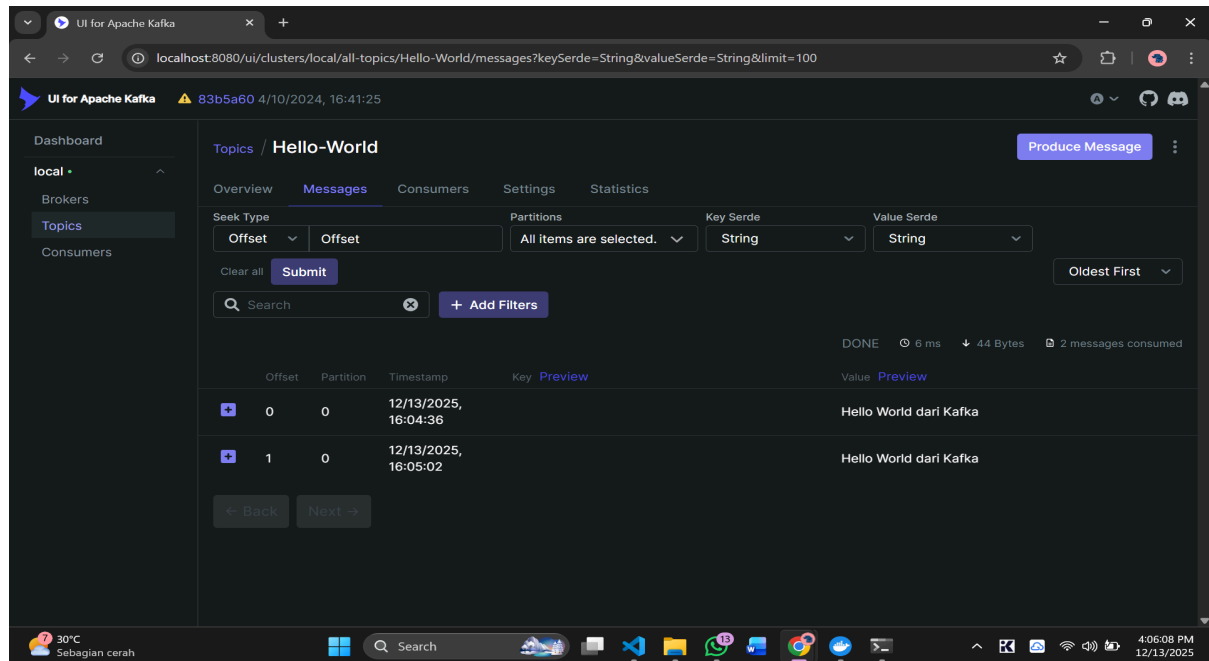
Langkah 5: Membuat *Topic* hello_world melalui Kafka UI

Dilakukan pembuatan *topic* Kafka menggunakan Kafka UI untuk pengujian apakah kafka sudah *terinstall* dengan baik, Kafka UI ini berjalan pada port yang telah dikonfigurasi. Kafka UI digunakan untuk memudahkan pengelolaan Kafka secara visual tanpa harus menggunakan perintah di terminal. Kafka UI diakses melalui browser dengan alamat: <http://localhost:8080>



Setelah *topic* berhasil dibuat, Kafka UI menampilkan informasi bahwa *topic* `hello_world` telah aktif dan siap digunakan. Keberhasilan pembuatan *topic* menandakan bahwa Kafka Broker berjalan dengan baik dan menampilkan pesan sesuai yang diinginkan.

contoh: “Hello World dari Kafka”.



Selain menggunakan Kafka UI, dapat digunakan terminal WSL untuk *running project topic*.

```
febri@FebriantiPuji: ~/kafka-
febri@FebriantiPuji:~/kafka-docker$ docker compose up -d
WARN[0000] /home/febri/kafka-docker/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 3/3
  ✓ Container zookeeper Running 0.0s
  ✓ Container kafka-ui Running 0.0s
  ✓ Container kafka Started 0.2s
febri@FebriantiPuji:~/kafka-docker$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
c8195490fb7b   provectuslabs/kafka-ui:latest       "/bin/sh -c 'java --..." 5 hours ago   Up 9 minutes  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
3ac35717a10d   confluentinc/cp-kafka:7.6.0         "/etc/confluent/dock..." 5 hours ago   Up 6 seconds  0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp
53d8de077c55   confluentinc/cp-zookeeper:7.6.0     "/etc/confluent/dock..." 5 hours ago   Up 9 minutes  0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp
181->2181/tcp   zookeeper
febri@FebriantiPuji:~/kafka-docker$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

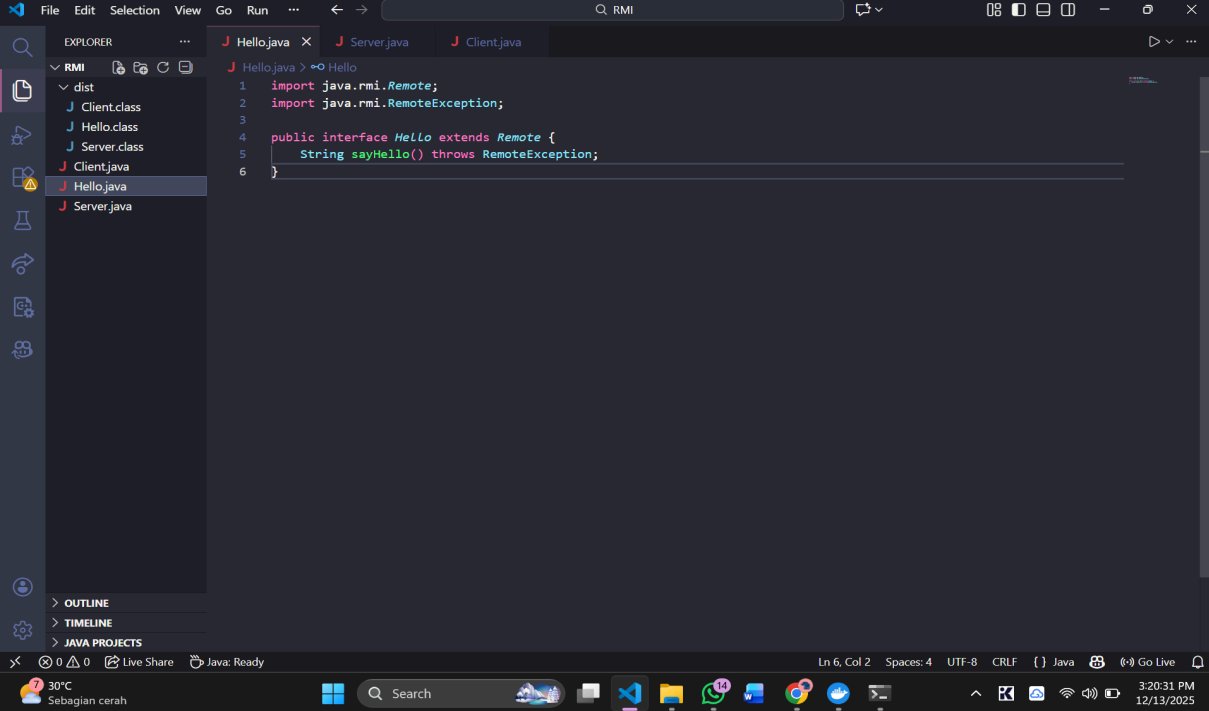
Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
```

3. Java RMI – Demo

Langkah 1: Membuat *Interface* Remote

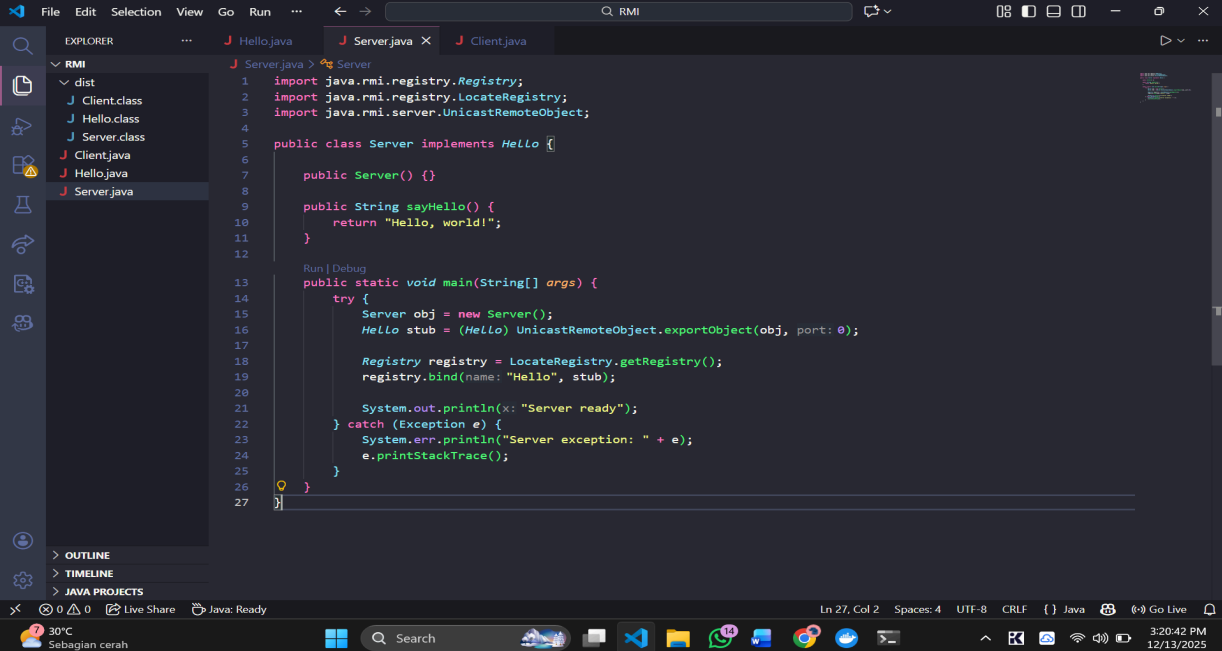
Interface remote mendefinisikan *method* yang dapat dipanggil oleh *client* secara remote.



```
1 import java.rmi.Remote;
2 import java.rmi.RemoteException;
3
4 public interface Hello extends Remote {
5     String sayHello() throws RemoteException;
6 }
```

Langkah 2: Implementasi *Server*

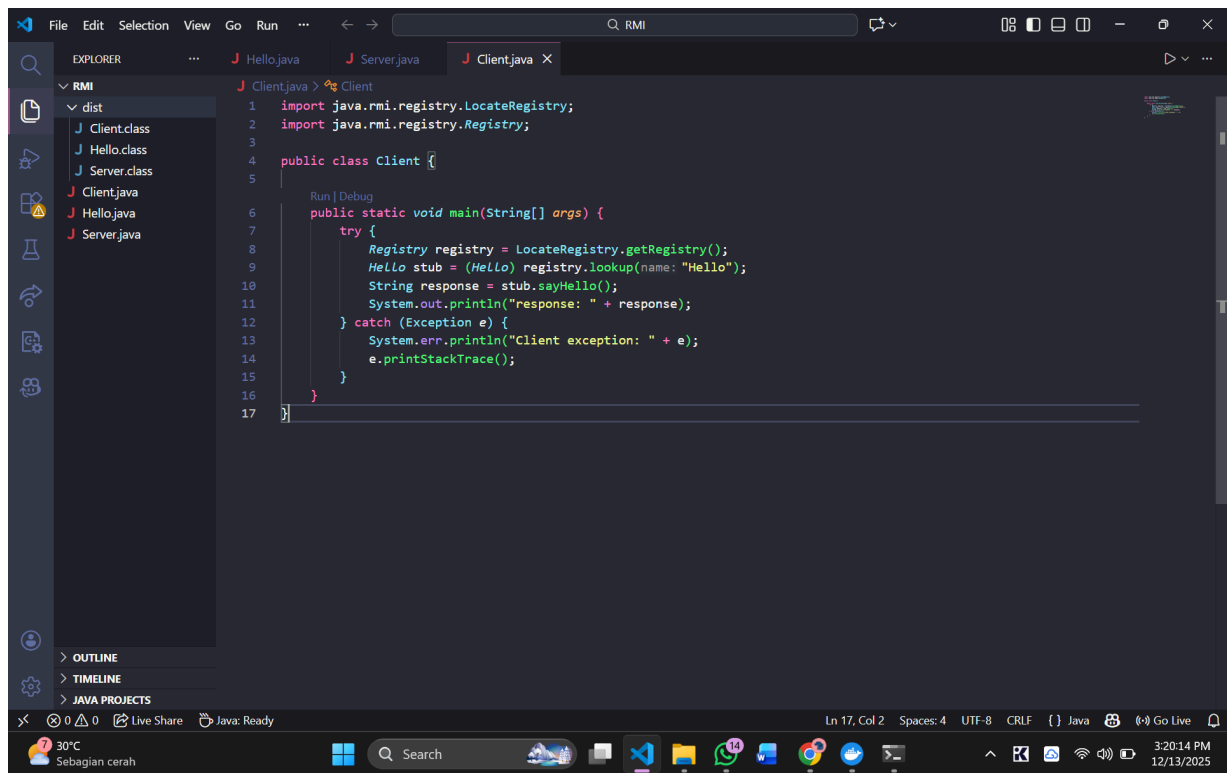
Server mengimplementasikan *interface* remote dan mendaftarkan *object* ke RMI Registry menggunakan nama tertentu.



```
1 import java.rmi.registry.Registry;
2 import java.rmi.registry.LocateRegistry;
3 import java.rmi.server.UnicastRemoteObject;
4
5 public class Server implements Hello {
6
7     public Server() {}
8
9     public String sayHello() {
10         return "Hello, world!";
11     }
12
13     public static void main(String[] args) {
14         try {
15             Server obj = new Server();
16             Hello stub = (Hello) UnicastRemoteObject.exportObject(obj, port: 0);
17
18             Registry registry = LocateRegistry.getRegistry();
19             registry.bind(name: "Hello", stub);
20
21             System.out.println(x: "Server ready");
22         } catch (Exception e) {
23             System.err.println("Server exception: " + e);
24             e.printStackTrace();
25         }
26     }
27 }
```

Langkah 3: Implementasi *Client*

Client digunakan untuk mengakses *object* remote yang telah didaftarkan pada RMI Registry dan memanggil *method* yang disediakan oleh *server*.



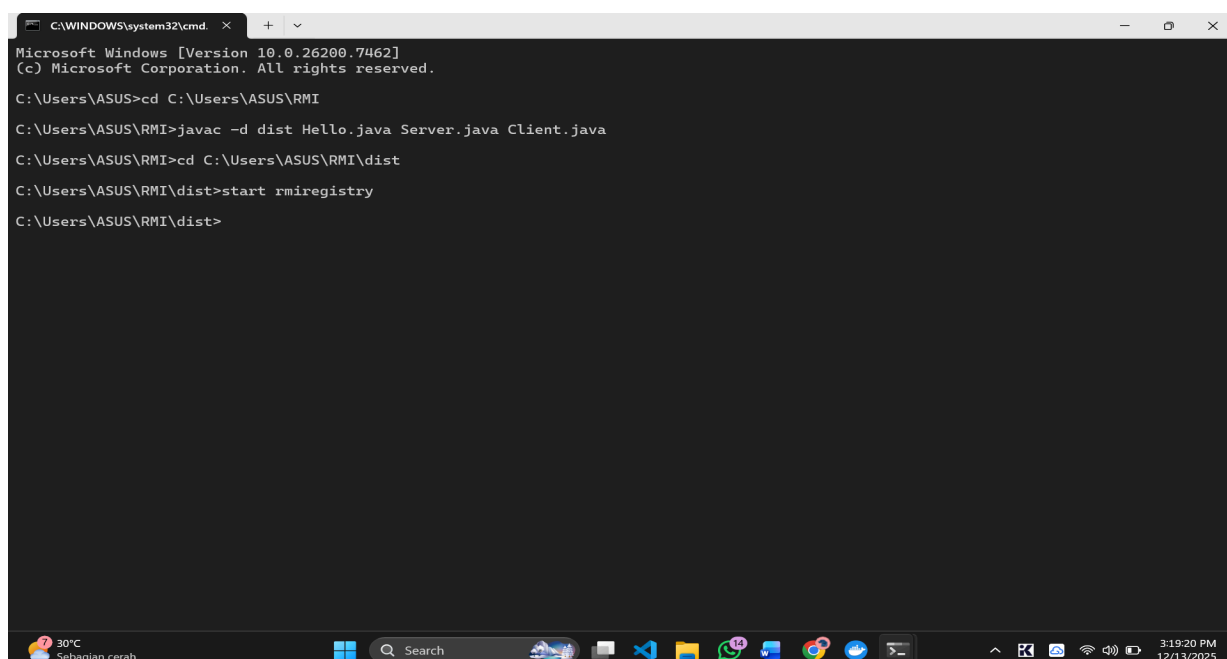
The screenshot shows an IDE with the following code in `Client.java`:

```
1 import java.rmi.registry.LocateRegistry;
2 import java.rmi.registry.Registry;
3
4 public class Client {
5
6     public static void main(String[] args) {
7         try {
8             Registry registry = LocateRegistry.getRegistry();
9             Hello stub = (Hello) registry.lookup(name: "Hello");
10            String response = stub.sayHello();
11            System.out.println("response: " + response);
12        } catch (Exception e) {
13            System.err.println("Client exception: " + e);
14            e.printStackTrace();
15        }
16    }
17 }
```

The Explorer panel on the left shows a package named `RMI` containing a sub-package `dist` with the following files: `Client.class`, `Hello.class`, `Server.class`, `Client.java`, `Hello.java`, and `Server.java`.

Langkah 4: Menjalankan RMI Registry

RMI Registry dijalankan menggunakan perintah `rmiregistry` sebagai tempat registrasi *object* remote. RMI Registry akan dijalankan pada CMD 1.

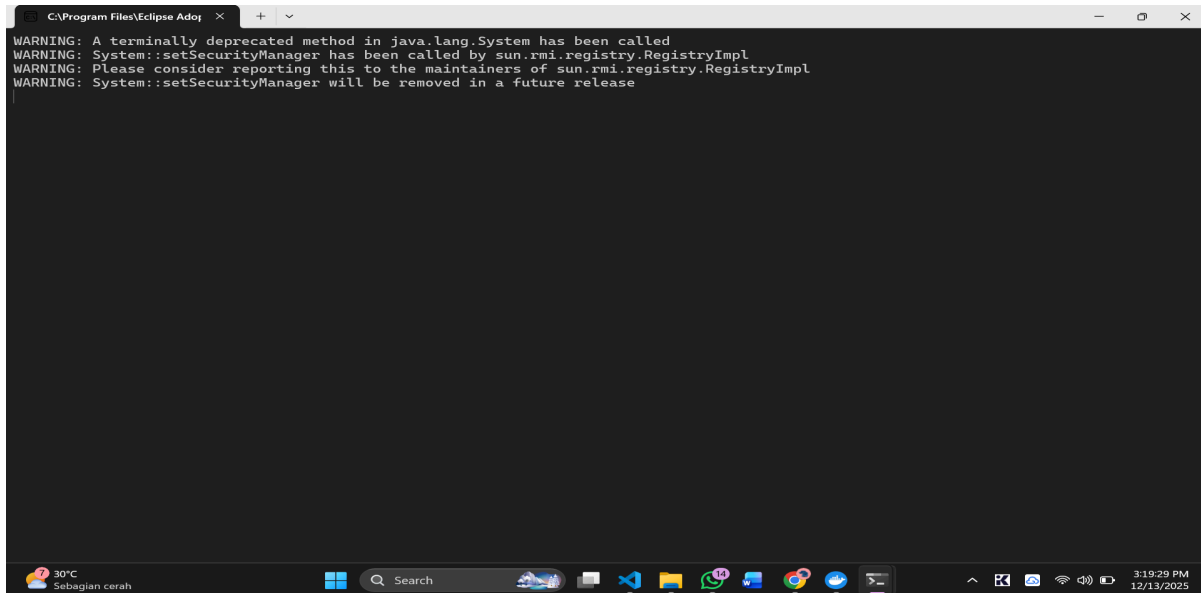


The screenshot shows a Windows Command Prompt with the following commands and output:

```
C:\WINDOWS\system32\cmd. x
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ASUS>cd C:\Users\ASUS\RMI
C:\Users\ASUS\RMI>javac -d dist Hello.java Server.java Client.java
C:\Users\ASUS\RMI>cd C:\Users\ASUS\RMI\dist
C:\Users\ASUS\RMI\dist>start rmiregistry
C:\Users\ASUS\RMI\dist>
```

Setelah menjalankan perintah `rmiregistry`, sistem membuka jendela terminal baru karena RMI Registry berjalan sebagai *service* terpisah yang harus tetap aktif. Registry berfungsi sebagai tempat pendaftaran *object* remote agar dapat diakses oleh *server* dan *client*. Pesan peringatan (*warning*) yang muncul merupakan peringatan dari Java versi terbaru dan tidak mempengaruhi jalannya aplikasi RMI.

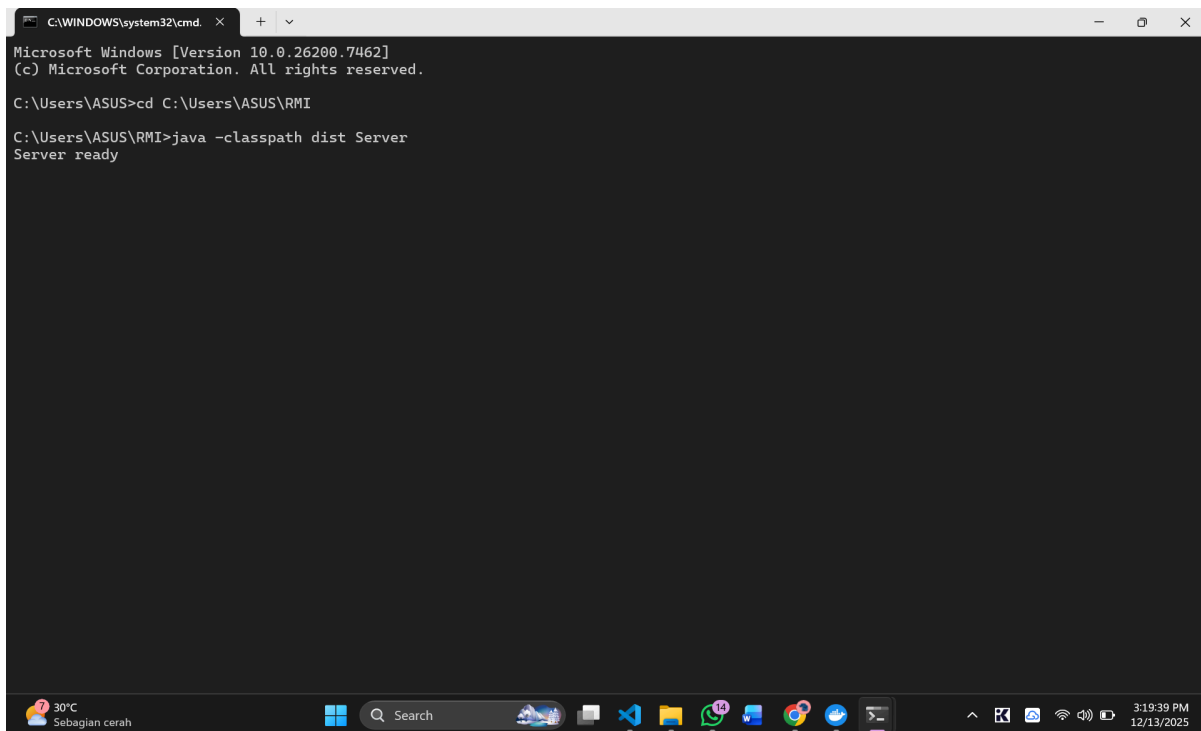


```
C:\Program Files\Eclipse Ado...
WARNING: A terminally deprecated method in java.lang.System has been called
WARNING: System::setSecurityManager has been called by sun.rmi.registry.RegistryImpl
WARNING: Please consider reporting this to the maintainers of sun.rmi.registry.RegistryImpl
WARNING: System::setSecurityManager will be removed in a future release
```

The screenshot shows a terminal window with Eclipse IDE warnings. The warnings are about deprecated methods in `java.lang.System` being called by `sun.rmi.registry.RegistryImpl`. The terminal is running on a Windows system, as indicated by the taskbar at the bottom showing the date and time as 3:19:29 PM on 12/13/2025.

Langkah 5: Menjalankan *Server*

Server dijalankan dan berhasil bind *object* ke registry. *Server* akan dijalankan pada CMD 2.



```
C:\WINDOWS\system32\cmd.
Microsoft Windows [Version 10.0.26200.7462]
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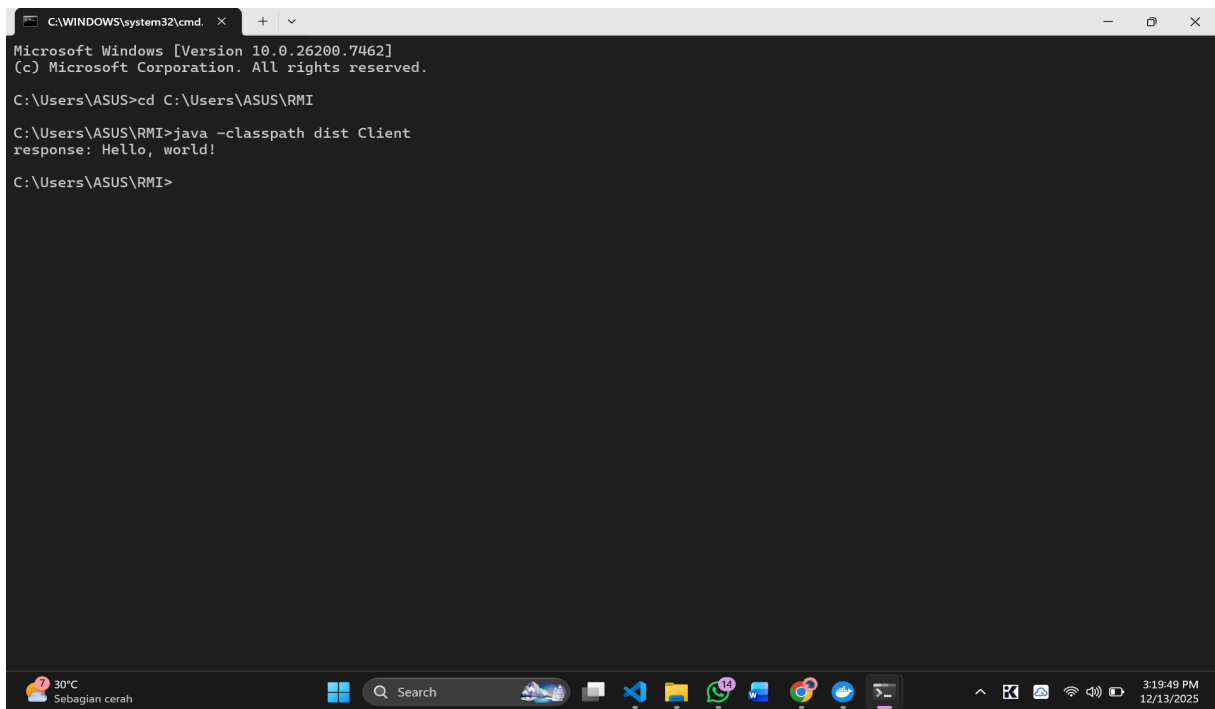
C:\Users\ASUS>cd C:\Users\ASUS\RMI

C:\Users\ASUS\RMI>java -classpath dist Server
Server ready
```

The screenshot shows a Windows Command Prompt window. The user has navigated to the directory `C:\Users\ASUS\RMI` and executed the command `java -classpath dist Server`. The output of the command is `Server ready`. The taskbar at the bottom shows the date and time as 3:19:39 PM on 12/13/2025.

Langkah 6: Menjalankan *Client*

Client memanggil *method* remote yang disediakan *server* dan menerima *response*. *Client* akan dijalankan pada CMD 3.



```
C:\WINDOWS\system32\cmd. X + v
Microsoft Windows [Version 10.0.26200.7462]
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C:\Users\ASUS>cd C:\Users\ASUS\RMI
C:\Users\ASUS\RMI>java -classpath dist Client
response: Hello, world!
C:\Users\ASUS\RMI>
```

The screenshot shows a Windows Command Prompt window with the following content: The title bar reads 'C:\WINDOWS\system32\cmd. X + v'. The window content shows the Windows version '10.0.26200.7462' and copyright information. The user navigates to 'C:\Users\ASUS\RMI' and runs 'java -classpath dist Client', which outputs 'response: Hello, world!'. The taskbar at the bottom shows the system clock as 3:19:49 PM on 12/13/2025, along with weather information (30°C, Sebagian cerah) and various application icons.

REFERENSI

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<https://kafka.apache.org/documentation/>
3. Oracle. (2024). *Java Remote Method Invocation (RMI)*.
<https://docs.oracle.com/javase/8/docs/technotes/guides/rmi/>
4. Oracle. (2024). *Java Platform, Standard Edition Documentation*.
<https://docs.oracle.com/javase/>

LAMPIRAN

Video Install Apache Kafka & JAVA RMI-Demo

<https://drive.google.com/file/d/1DN913Sp8MJl9y0eFN5UixW7FxjVOlgl1/view?usp=drivesdk>